

PLA vs VIDEO EXTENSOMETER - MOT SESSION PACK

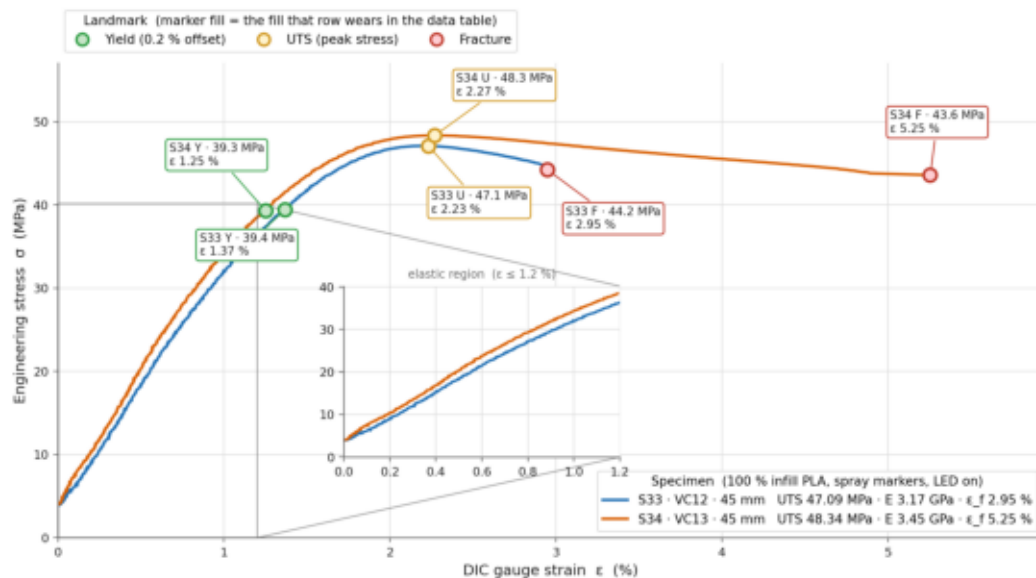
What to bring, what must match, and what we are comparing against

WHAT WE ARE VALIDATING. Our strain comes from two spray dots tracked in a Basler frame, 45 mm apart, at about 20.9 px/mm. Everything the campaign claims about PLA rests on that measurement being right. A video extensometer is an independent optical measurement of the same quantity, so agreement is the strongest evidence available without a contact extensometer.

WHAT WE BRING TO COMPARE. 13 fractured 100 % infill PLA specimens: UTS 46.75 +/- 0.99 MPa (CV 2.1 %), E 3.06 +/- 0.27 GPa (CV 8.7 %). Page 3 has the full table.

THE ONE THING THAT WILL COST YOU THE SESSION: THE STRAIN ZERO. Our Px0 is frozen AFTER a ~300 N preload, so our strain axis begins at an already-seated specimen. An extensometer zeroed at 0 N carries the toe - grip take-up and specimen straightening - inside its strain, which LOWERS apparent modulus and shifts the 0.2 % offset yield. That difference is bigger than the 8.5 % scatter of our entire 11-specimen set. Match it, or agree in advance to correct for it. Do not discover it in the data.

THE TWO RUNS CAPTURED FOR THIS COMPARISON - S33 AND S34



CONTENTS

- Parameters that must match - the pre-flight
- Our reference values - the 100 % PLA set
- Comparison sheet - fill in at the bench
- Traps that make two correct instruments disagree
- What to physically bring

START WITH OUR OWN FRAMES

We have every frame of S33 and S34 as PNG stills plus AVI - dropped, 19.9 fps, 100 % DIC coverage. Running OUR frames through THEIR software is the cleanest validation there is: identical pixels, two independent algorithms, no specimen, no machine and no operator in the difference. Bring them on a stick and ask before mounting anything.

If that is not possible, the fallback is a live side-by-side on one specimen, which is still worth doing but now carries their machine's compliance and their grips in the comparison as well as their optics.

PARAMETERS THAT MUST MATCH

Agree every row BEFORE the first pull. Right-hand column is theirs to fill in.

Parameter	Our value	Why it matters	MOT setting
Gauge length	80.0 mm	Marker-to-marker, not the specimen's parallel length	
Marker spacing at zero	~1674 px @ 20.93 px/mm	Our Px0. Their targets should sit on the SAME dots	
Cross-section	80.0 mm ²	CAD nominal, NOT measured. See the infill note on p5	
Infill	100 %	Solid. Our 50 % set needs a separate knock-down factor	
Stress convention	Engineering	Force / original area. We do not report Cauchy	
Strain convention	Engineering dL/L0	CSV also carries a true-strain column; ignore it	
STRAIN ZERO	AFTER ~300 N preload	THE BIG ONE - see the banner below	
Crosshead speed	0.10 mm/s	Constant. Not a strain-rate-controlled ramp	
Modulus fit window	Steepest straight run	NOT a fixed window - see p5	
Yield definition	0.2 % offset	Offset line has slope E, so it moves if E does	
Sampling	DIC 19.9 Hz / load ~11 Hz	Ask theirs; a slow extensometer smooths the knee	
Temperature / RH	record it	PLA Tg is 55-60 degC; room conditions matter	

STRAIN ZERO, IN NUMBERS: at 300 N our specimen is already carrying 3.75 MPa. If their extensometer zeroes at 0 N, its first per cent of strain contains grip take-up that ours never saw, and every modulus they quote will read LOW against ours for a reason that has nothing to do with either instrument.

IF THEY CANNOT PRELOAD TO 300 N: let them zero at 0 N, but then compare on the SECANT between two strain points that both lie above the toe - 0.05 % and 0.25 % is the ISO 527 pair - rather than on either instrument's own automatic modulus. That comparison is immune to where the axes were zeroed. Record BOTH numbers either way.

OUR REFERENCE VALUES - 100 % INFILL PLA

From registry.json at build time. Every row is a specimen that fractured.

Specimen	Date	UTS MPa	sy MPa	E GPa	ef %	anchor N	
S7	2026-06-17	47.82	45.42	3.003	2.98	475	Mk I
S8	2026-06-25	44.83	42.99	2.686	5.17	476	Mk I
S10	2026-06-25	46.82	44.39	2.896	7.32	470	Mk I
S11	2026-06-25	46.10	43.72	2.886	5.00	470	Mk I
S9	2026-06-25	45.47	42.30	3.075	7.44	475	Mk I
S16	2026-07-28	47.40	44.10	2.638	6.00	418	Mk I
S24	2026-08-14	46.47	39.46	3.004	7.44	307	Mk II
S25	2026-08-17	46.16	39.82	3.167	4.18	280	Mk II
S26	2026-08-17	47.17	41.75	3.549	5.74	308	Mk II
S12	2026-08-17	47.78	41.99	3.286	5.82	317	Mk II
S13	2026-08-18	46.33	42.07	3.030	5.15	307	Mk II
S33	2026-08-24	47.09	39.43	3.174	2.95	310	<< EXTENSOMETER PAIR
S34	2026-08-24	48.34	39.31	3.451	5.25	318	<< EXTENSOMETER PAIR

AMBER ROWS - S33 and S34 - ARE THE COMPARISON PAIR. Both were run with frame capture expressly for extensometer cross-validation. Every frame still exists, so their software can re-process our pixels rather than test a new specimen. Quote these two first.

GREEN ROWS are PPD-UTM Mk II — after the 2026-08-12 realignment. Prefer them if a wider subset has to be quoted: same rig, same DIC, and a load path known to be free of the binding that produced the earlier stalls.

eps_f scatters at about 25 % and always has. It is the value most sensitive to where fracture is declared and to dropout rows near the end of a run, so treat a disagreement in eps_f as far weaker evidence than one in E or UTS.

	mean	sd	CV	range	
UTS	46.75	0.99	2.1 %	44.83 - 48.34	MPa
sigma_y	42.06	2.06	4.9 %	39.31 - 45.42	MPa
E	3.06	0.27	8.7 %	2.64 - 3.55	GPa
eps_f	5.42	1.48	27.3 %	2.95 - 7.44	%

AGAINST THE add:north E-PLA TDS

	TDS	PPD-UTM	of TDS
UTS	58.0	46.8	81 %
E	2.87	3.06	107 %
eps_f	8	5.4	68 %

We land near the datasheet on STIFFNESS and about a fifth below on STRENGTH. That gap is the expected print knock-down - layer adhesion, not instrument error - and it is exactly what an independent extensometer is being asked to confirm or refute.

COMPARISON SHEET

Fill in at the bench. One block per specimen. Record BOTH raw numbers, never just the difference.

Specimen 1 ID _____ date _____ area _____ mm² gauge _____ mm strain zero at _____ N speed _____ mm/s

Property	PPD-UTM (DIC)	XT-205 (extensometer)	difference %	notes
E GPa				
sigma_y MPa				
UTS MPa				
eps_f %				
secant 0.05-0.25 % GPa				

Specimen 2 ID _____ date _____ area _____ mm² gauge _____ mm strain zero at _____ N speed _____ mm/s

Property	PPD-UTM (DIC)	XT-205 (extensometer)	difference %	notes
E GPa				
sigma_y MPa				
UTS MPa				
eps_f %				
secant 0.05-0.25 % GPa				

Specimen 3 ID _____ date _____ area _____ mm² gauge _____ mm strain zero at _____ N speed _____ mm/s

Property	PPD-UTM (DIC)	XT-205 (extensometer)	difference %	notes
E GPa				
sigma_y MPa				
UTS MPa				
eps_f %				
secant 0.05-0.25 % GPa				

Write down the RAW numbers from both instruments before computing any difference. A difference recorded without its two inputs cannot be re-checked once you have left.

TRAPS - HOW TWO CORRECT INSTRUMENTS DISAGREE

Each of these has already moved one of our numbers. Check them before concluding anything about either instrument.

1. Strain zero

The PPD-UTM starts after a ~300 N preload; theirs probably starts at 0 N. The toe region is then inside their strain and not ours. Lowers their apparent E and moves the 0.2 % offset yield with it. Fix: match the preload, or compare on the 0.05-0.25 % secant, which does not care where zero was.

3. Area basis

80 mm² is the CAD nominal, not a measured section. A printed part is usually slightly under nominal, so measure the actual section at MOT with their calipers and record it. Stress scales inversely, so a 3 % area error is a 3 % UTS error.

5. Dropout rows

Rows where DIC found no markers carry $ec = 0.0$, not a gap. Averaging them in drags strain toward zero. Our analysis drops them; if you export raw CSV for their software, filter `DIC_Blobs = 2` first.

2. Modulus fit window

We report the STEEPEST STRAIGHT RUN, chosen per specimen, not a fixed strain window. Against a fixed 0.05-0.40 % window the same runs read 2.67 GPa instead of 3.02, and the scatter doubles. If their software uses a fixed window, ask for the window and compare like for like - we can recompute ours either way from the raw CSV.

4. Infill is not a material

These are 100 % infill so the nominal area is defensible. Do NOT let a 50 % specimen into this comparison - it needs a knock-down factor of about 2.4 and would look like an instrument fault.

6. Post-fracture jump

After fracture the markers fly apart and `L_px` jumps. One such row is enough to ruin `eps_f`. The app now refuses them at source, but any older CSV you hand over may contain them - trim to the fracture index.

If ours and theirs agree within a few per cent on E and UTS, the DIC measurement is validated and the campaign's PLA numbers stand. If they do not, work down this list before doubting either instrument.

WHAT TO PHYSICALLY BRING

Tick before leaving.

PRINTED

- ☐ This pack (all 6 pages)
- ☐ S33_S34_stress_strain_reference.pdf - stress at every 0.10 % strain step, so a number read off their software can be looked up rather than eyeballed off a curve
- ☐ Per-specimen report PDFs for S33 and S34
- ☐ add:north E-PLA technical data sheet rev 2.1
- ☐ E_modulus_explained.pdf - why our fit window is what it is

PHYSICAL

- ☐ Spare printed specimens, 100 % infill, same spool and slicer settings - bring more than you think you need
- ☐ The spray paint used for the dots, in case their optics want fresh targets
- ☐ Calipers, to measure the actual cross-section on site
- ☐ The spool label or a photo of it - brand, batch, diameter

ON A USB STICK

- ☐ S33 and S34 frame folders - PNGs plus the AVIs. The single most valuable item here if their software can ingest video
- ☐ Both raw CSVs, unfiltered
- ☐ registry.json - the whole specimen register
- ☐ This pack as a PDF, in case a page is needed on screen

KNOW BEFORE YOU GO

- ☐ Their load cell range and its calibration date
- ☐ Whether their extensometer tracks our dots or needs its own targets
- ☐ Whether they can preload to 300 N before zeroing strain
- ☐ Their modulus fit window and yield definition
- ☐ Their sampling rate

Ask for their raw data before you leave the building. A comparison you cannot recompute later is a comparison you have to trust from memory.